

Experiment 1

Data Acquisition and Data Profiling using BI Tools

Aim

To acquire data from multiple sources and perform data profiling using Microsoft Power BI in order to assess data quality, understand data structure, identify anomalies, and prepare data for further analytical processing.

Tools / Software Required

- Microsoft Power BI Desktop
- Sample datasets: Sales_Data.csv, Employee_DB.xlsx, and a sample SQL database

Theory

Business Intelligence (BI)

Business Intelligence refers to a set of technologies, processes, and practices that enable organizations to collect, integrate, analyse, and present business information. BI helps transform raw data into meaningful, actionable insights that support better decision-making across an organization.

Key components of a BI system include:

- Data Sources: Databases (SQL/NoSQL), flat files, cloud stores, and APIs
- ETL Pipeline: Extract, Transform, Load processes that move data from source to destination
- Data Warehouse / Data Lake: Centralized repositories for structured or unstructured data
- BI & Reporting Tool: Microsoft Power BI for data connection, transformation, and visualization
- Analytics Layer: Descriptive, diagnostic, predictive, and prescriptive analytics

Data Acquisition

Data Acquisition is the process of collecting and importing data from various heterogeneous sources into a unified environment for processing. It is the first and foundational step in any BI pipeline. In Power BI, data is acquired through the Get Data option, which supports a wide range of connectors including flat files (CSV, XLSX), relational databases (SQL Server, MySQL), cloud sources (SharePoint, Azure), and web-based sources.

Data Profiling

Data Profiling is the process of examining data from an existing source and collecting statistics, summaries, and metadata about that data. It is a critical quality assurance step that reveals the

structure, content, and quality of a dataset before further processing. Data profiling covers the following key dimensions:

- Completeness: Percentage of non-null values in each column.
- Uniqueness: Ratio of distinct to total values — detects duplicate records.
- Validity: Whether values conform to defined rules or expected formats.
- Consistency: Uniformity of values across related datasets.
- Accuracy: Correctness of values relative to real-world facts.
- Timeliness: Currency of data relative to the reporting period.

Data Profiling in Power BI — Power Query Editor

Microsoft Power BI provides built-in data profiling capabilities through the Power Query Editor. Under the View tab, three profiling features can be enabled:

- Column Quality: Displays the percentage of valid, error, and empty values per column.
- Column Distribution: Shows a histogram of value frequency along with the distinct and unique count for each column.
- Column Profile: Provides detailed statistics for a selected column including minimum, maximum, mean, median, standard deviation, and a full value distribution chart.

Conduct

Part A: Data Acquisition

Step 1: Launch Power BI Desktop and import the CSV file.

```
Home > Get Data > Text/CSV  
Browse to Sales_Data.csv > Load
```

Step

2:

Import

Sales_Data.csv

File Origin

Delimiter

Data Type Detection

1252: Western European (Windows) ▾

Comma ▾

Based on first 200 rows ▾

SaleID	Date	CustomerID	Product	Quantity	UnitPrice	SalesAmount	Region
1	5/21/2025	C1031	Mouse	1	25	25	South
2	2/14/2025	C1075	Monitor	1	500	500	North
3	4/22/2025	C1029	Printer	1	25	25	North
4	11/29/2025	C1089	Printer	5	50	250	West
5	10/29/2025	C1035	Laptop	2	250	500	South
6	5/23/2025		Mouse	4	100	400	East
7	7/14/2025	C1012	Keyboard	1	25	25	East
8	1/23/2025	C1093	Monitor	5	100	500	North
9	10/10/2025	C1037	Printer	4	25	100	East
10	12/27/2025	C1008	Laptop	5	50	250	South
11	4/30/2025	C1012	Monitor	3	25	75	East
12	3/25/2025	C1047	Keyboard	4	100	400	South
13	11/8/2025	C1081	Mouse	3	25	75	South
14	7/14/2025	C1034	Printer	2	250	500	South
15	4/28/2025	C1004	Keyboard	3	25	75	West
16	4/19/2025	C1072	Keyboard	3	25	75	South
17	11/26/2025	C1058	Mouse	4	250	1000	East
18	10/15/2025	C1068	Keyboard	2	50	100	West
19	7/5/2025	C1028	Mouse	5	250	1250	West
20	2/26/2025	C1019	Mouse	1	25	25	West

The data in the preview has been truncated due to size limits.

Extract Table Using Examples

Load

Transform Data

Cancel

the Excel file.

Home > Get Data > Excel Workbook

Browse to Employee DB.xlsx > Select the appropriate sheet > Load

SaleID	Date	CustomerID	Product	Quantity	UnitPrice	SalesAmount	Region
1	Wednesday, May 21, 2025	C1031	Mouse	1	25	25	South
2	Friday, February 14, 2025	C1075	Monitor	1	500	500	North
3	Tuesday, April 22, 2025	C1029	Printer	1	25	25	North
4	Saturday, November 29, 2025	C1089	Printer	5	50	250	West
5	Wednesday, October 29, 2025	C1035	Laptop	2	250	500	South
6	Friday, May 23, 2025		Mouse	4	100	400	East
7	Monday, July 14, 2025	C1012	Keyboard	1	25	25	East
8	Thursday, January 23, 2025	C1093	Monitor	5	100	500	North
9	Friday, October 10, 2025	C1037	Printer	4	25	100	East
10	Saturday, December 27, 2025	C1008	Laptop	5	50	250	South
11	Wednesday, April 30, 2025	C1012	Monitor	3	25	75	East
12	Tuesday, March 25, 2025	C1047	Keyboard	4	100	400	South
13	Saturday, November 8, 2025	C1081	Mouse	3	25	75	South
14	Monday, July 14, 2025	C1034	Printer	2	250	500	South
15	Monday, April 28, 2025	C1004	Keyboard	3	25	75	West
16	Saturday, April 19, 2025	C1072	Keyboard	3	25	75	South
17	Wednesday, November 26, 2025	C1058	Mouse	4	250	1000	East
18	Wednesday, October 15, 2025	C1068	Keyboard	2	50	100	West
19	Saturday, July 5, 2025	C1028	Mouse	5	250	1250	West
20	Wednesday, February 26, 2025	C1019	Mouse	1	25	25	West
21	Thursday, July 17, 2025	C1048	Printer	5	25	125	West
22	Saturday, October 11, 2025	C1001	Laptop	5	100	500	East
23	Saturday, May 31, 2025	C1055	Mouse	3	25	75	West
24	Sunday, September 14, 2025	C1097	Mouse	1	100	100	North
25	Saturday, November 8, 2025	C1025	Mouse	2	500	1500	East

Step 3: Connect to the SQL database and import the Transactions table.

Home > Get Data > SQL Server

Enter server name and database credentials > Select Transactions table > Load

SQL Server database

Server ①
Transaction.sql

Database (optional)

Data Connectivity mode ①
☒ Import
☐ DirectQuery

▸ Advanced options

OK Cancel

Verify that all three data sources appear in the Data pane on the right-hand side of the Power BI window.

Part B: Data Profiling in Power Query Editor

Step 4: Open Power Query Editor.

Home > Transform Data

Queries [1] < X fx = Table.TransformColumnTypes(#"Promoted Headers",{{"SaleID", Int64.type}, {"Date", type date},

	SaleID	Date	CustomerID	Product	Quantity
1	1	5/21/2025	C1031	Mouse	
2	2	2/14/2025	C1075	Monitor	
3	3	4/22/2025	C1029	Printer	
4	4	11/29/2025	C1089	Printer	
5	5	10/29/2025	C1035	Laptop	
6	6	5/23/2025		Mouse	
7	7	7/14/2025	C1012	Keyboard	
8	8	1/23/2025	C1093	Monitor	
9	9	10/10/2025	C1037	Printer	
10	10	12/2/2025	C1008	Laptop	
11	11	4/30/2025	C1012	Monitor	
12	12	3/29/2025	C1047	Keyboard	
13	13	11/8/2025	C1081	Mouse	
14	14	7/14/2025	C1034	Printer	
15	15	4/28/2025	C1004	Keyboard	
16	16	4/19/2025	C1072	Keyboard	
17	17	11/26/2025	C1058	Mouse	
18	18	10/15/2025	C1068	Keyboard	
19	19	7/5/2025	C1028	Mouse	
20	20	2/26/2025	C1019	Mouse	
21	21	7/17/2025	C1048	Printer	
22	22	10/11/2025	C1001	Laptop	
23	23	5/31/2025	C1055	Mouse	
24	24	9/14/2025	C1097	Mouse	
25					

Query Settings

PROPERTIES

Name
Sales_Data

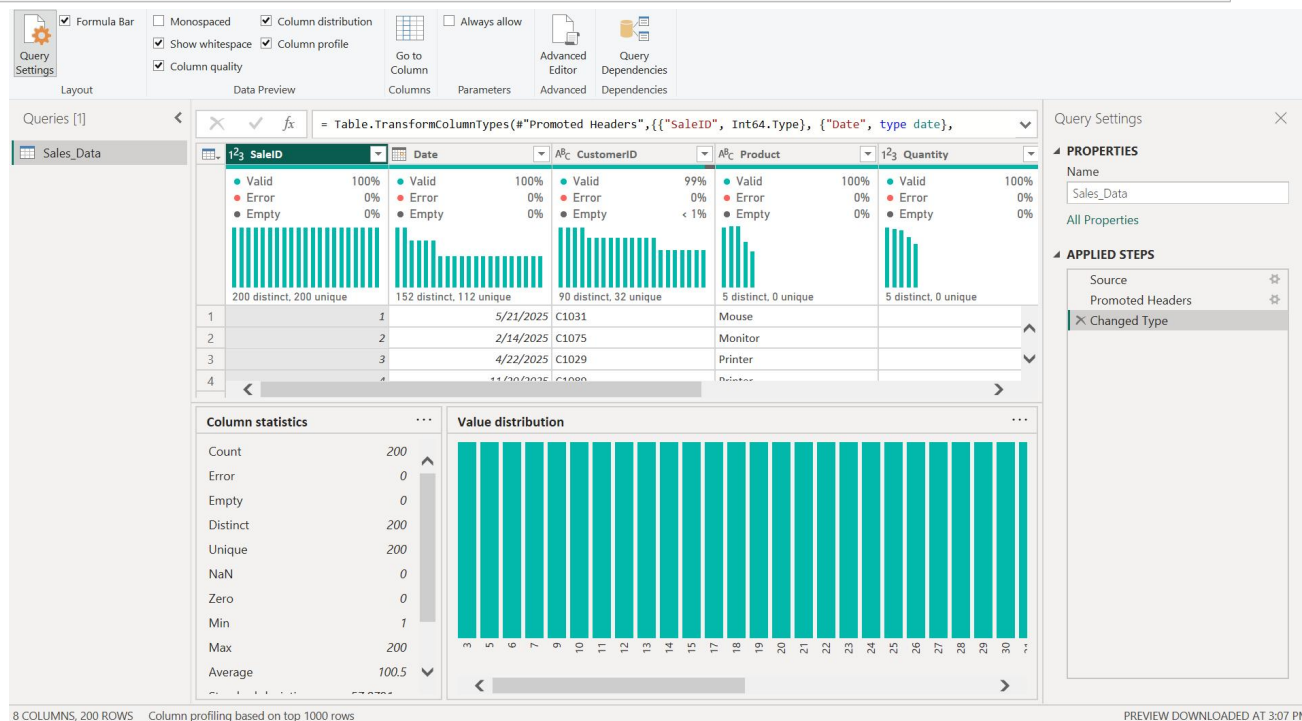
All Properties

APPLIED STEPS

Source
Promoted Headers
X Changed Type

Step 5: Enable data profiling features.

```
View tab > tick: Column Quality
View tab > tick: Column Distribution
View tab > tick: Column Profile
```

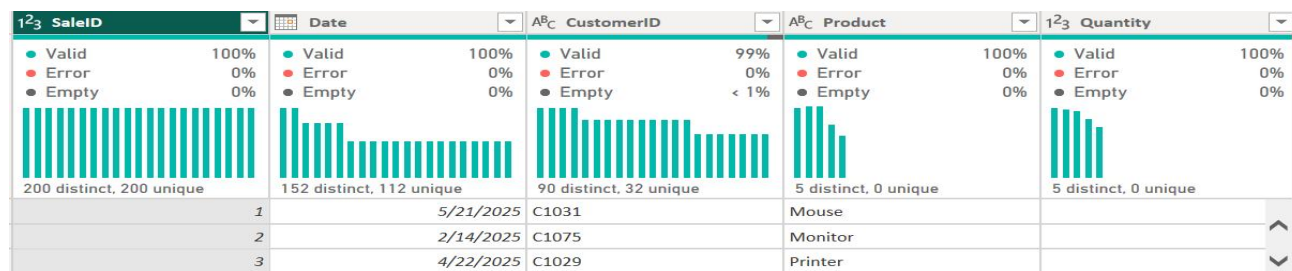


Select each column one at a time and observe the profiling statistics displayed below the data preview grid.

Step 6: Record profiling observations.

For each column in each dataset, note the following:

- Data type and total row count
- Null / empty percentage (from Column Quality)
- Distinct value count and unique value count (from Column Distribution)
- Minimum, maximum, mean, and standard deviation for numeric columns (from Column Profile)
- Any errors, mismatched data types, or unexpected values observed

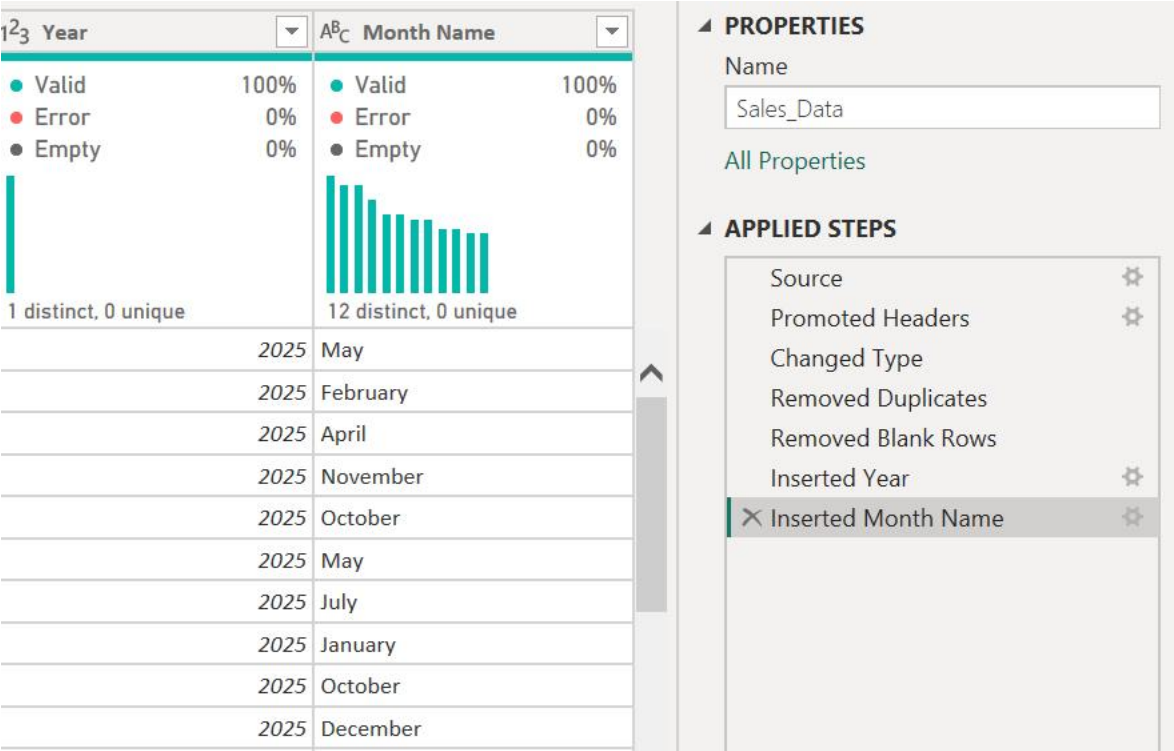


Part C: Identify and Document Data Quality Issues

Step 7: Based on the profiling results, identify data quality issues.

Look for the following common issues:

- High null percentage in important columns such as customer ID or date fields
- Columns with only one unique value — zero variance and no analytical value
- Numeric columns containing text values indicating a data type mismatch
- Date columns with future dates or dates outside the expected range
- Duplicate rows identified through a low uniqueness ratio



Document at least three data quality issues found and note the suggested remediation action for each.

Result

Data from multiple heterogeneous sources (CSV, Excel, and SQL database) was successfully acquired and imported into Microsoft Power BI Desktop. Data profiling was performed using the Power Query Editor's built-in tools — Column Quality, Column Distribution, and Column Profile. Key data quality dimensions including completeness, uniqueness, and validity were assessed across all datasets, and data quality issues were identified and documented for remediation prior to analysis.

Learning Outcomes

After completing this experiment, students will be able to:

- Explain the concept of Business Intelligence, its key components, and the role of data acquisition in a BI pipeline. (CO1 - BT2)
- Describe different data source types and the methods used to connect and import them into Power BI. (CO1 - BT2)
- Connect to multiple heterogeneous data sources and successfully import datasets using Power BI's Get Data feature. (CO2 - BT3)
- Apply data profiling techniques in Power Query Editor to assess data quality dimensions across real-world datasets. (CO2 - BT3)
- Interpret profiling statistics to identify data quality issues and propose suitable remediation strategies. (CO2 - BT3)